PHONATION

At the end of this section, the students will be able to:

- EXPLAIN various aspects of phonation and states of the glottis during the proc sounds.

Topic-095: Introduction to Phonation

The position of larynx (also known as sound box) and the vocal folds important in the description of speech sounds. 'Phonation' is a technical term forms of vibration of the vocal folds (or vocal cords) and the process is mor voicing. The glottis (which is defined as the space between the vocal folds) c shapes (such as voiced, voiceless, murmuring and creaky positions). The most co folds are used to describe the two possible features of consonant sounds by c either voiceless with the vocal folds apart (such as /p/ and /t/) or voiced with the that they will vibrate when air passes between them (such as /b/ and /g/). important in the description of speech sounds in particular languages an pathological voices. The process of phonation is also known as 'voicing' and lary

Topic-096: Phonation Explanation

'Phonation' is a general term used in phonetics to refer to any vocal a sound box). The possibilities of various kinds of vocal-fold vibration (voicing c phonatory activities, and the study of phonation types is aimed at accounting possibilities (such as voiced, voiceless, breathy and creaky voice). Some pho modifications in phonation under this heading which stem from variations in ler of the vocal folds as displayed in the various registers of speech. Actually dur what happens is that the air passes between the vocal folds and modification to in the laryngeal area thus creating variation in intensity, frequency and quality important role in speech including the voicing of sounds and murmuring.

Topic-097: States of the Glottis

As discussed earlier, the space between the vocal folds (glottis - insic number of positions thus modifying the form of speech sounds and changi brought into light this contact of vocal folds with each other, the shape of forced to produce phonation or voicing. So, based on the possibilities of the n of glottis are determined thus the vibration can be made to vary in many way such things as pitch, loudness and voice quality. If a narrow opening is ma friction noise can result and this is found in whispering and in the glottal

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open glottis is found in most voiceless consonants. In Table 6.6 (chapter six o possible states of the glottis are given. Go through them and explore further.

Topic-098: Phonation Types

In order to further explain the nature of the phonation as a process, we phonation types and its description. There are mainly four possible glottis/l phonation:

- Voiceless – when the folds are open apart and the air passing through t
- Voiced – when the folds are tight together and there is vibration durin glottis (e.g., /b/ or /d/).
- Creaky voice – when there is a slight opening in the front and the a together, so that the vocal folds can vibrate only at the anterior end (th
- Breathy or murmuring sound – when the vocal folds are apart but breathy voice is like a whisper except voice.

Topic-099: Voicing and Consonants

Voicing is an important feature of speech sounds which is used not c istinction for describing sounds. There are sounds (such as vowels, n onorants) which are usually voiced (though in particular contexts the voic n the other hand, there are sounds which may be voiced or voiceless su ogether called obstruents). Obstruents are the most frequently found soun oicelessness. Remember that voiced and voiceless can make a phonemic c English) thus the meaning of a word can change if one uses a voicele und. A glottal stop is only an allophonic variation and is used in RP be ese three can even be replaced by a glottal stop altogether. The symb estion mark [ʔ].

urther readings on this section:

hapter 6 of the textbook (**A Course in Phonetics** by Peter Ladefoged an hapter 4 of the additional reading book (**English Phonetics and Phonol** eter Roach)
nline sources: http://renu-voice.com/my-voice/
https://mybookfile.wordpress.com/2016/01/02/phonetics

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VOICE ONSET TIME (VOT)

e end of this section, the students will be able to:

XPLAIN and think about analyzing VOT in various languages.

-100: Voice Onset Time (VOT)

All human languages distinguish between voiced and voiceless consonants, an most common consonants to be distinguished using the voicing feature. Howe matter of a plosive being either completely voiced or completely voiceless. T in relation to the consonant articulation is also very important. In one particul ble that it has for a long time been given its own name: aspiration, in which the does not happen until for some time after the release of the plosive (usually voic , has been the subject of much experimental investigation which has led to the ic measure of voice timing called voice onset time (or VOT). In simple words, the sive may lag behind the plosive release, or it may precede ("lead") contrarily, re ally voiced plosive. Both can be represented on the VOT scale, one case havin other negative values (and the third possibility is zero VOT).

01: VOT Explanation

Voice Onset Time (VOT) is a term used in phonetics referring to the point in time ation starts in relation to the release of a closure (during the production of plo understand VOT, the three types of plosive sounds are to be explained – voiced aspirated sound. For example, during the production of a fully voiced plosive folds vibrate throughout; in a voiceless unaspirated plosive (such as /p/ or /t/), before voicing starts; and, in a voiceless aspirated plosive (e.g., /p^h/ or /t^h/), th epending on the amount of aspiration. The amount of this delay is called Vo hich in relation to the types of plosive varies from language to language.

2: Importance of VOT

OT is an important feature in experimental phonology and it is used in orde different languages and their stop sounds. Languages vary in terms of VOT and an important feature to be explored for the comparison of languages. It al insight regarding the perception of VOT by bilingual learners. Language speci important to be considered by language experts. The VOT values can also prov he phonemic contrast of appropriate sound production by the learners. Moreov egarding the three different possibilities (types of VOTs). VOT is calculated thr gy and important contrastive features are to be taken into account.

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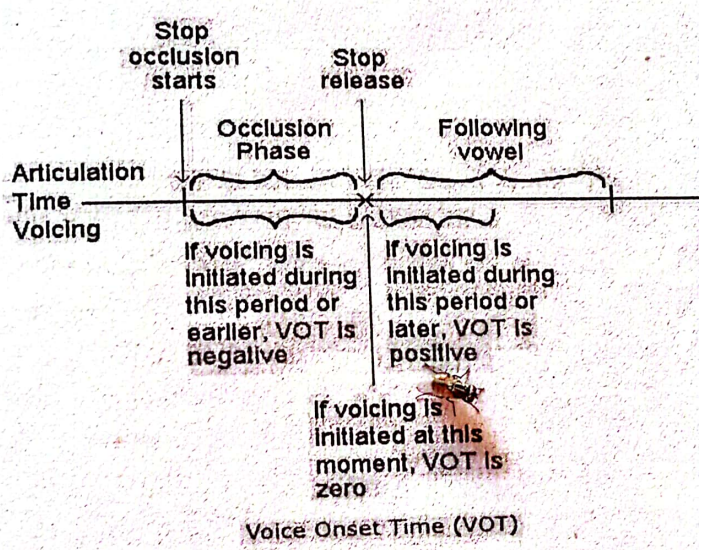
Remember that different languages choose different points along the VOT c... positions among stop consonants. Different possibilities that occur in different l... with reference to a scale going from most aspirated (largest positive VOT) at the ...gest negative VOT) at the bottom. The Navajo aspirated stops have a very larg... ...ce exceptional (150 MS). On the other hand, the normal value for the VOT of E... ...would be between 50 and 60 MS.

Topic-103. Types of VOT

There are three possible types of VOT based on the nature of stop s... ...aspirated voiceless stops have a voice onset time at or near zero. This means ...following vowel begins at or near to when the stop is released. The second possib... ...stops are followed by a vowel: voice onset time is greater than zero called a positi... ...the VOT in such cases is based on the practical measure of aspiration – the longer ...the aspiration (Navajo, for example, has strongly aspirated stops where, the aspira... ...VOT) lasts twice as long as that of English - 150ms). The third possibility is wh... ...VOT noticeably less than zero called "negative VOT". This would simply mean th... ...vibrating before the stop is released. Analyze the following ... (http://clas.mq.edu.au/speech /phonetics/phonetics/airstream_laryngeal/vot.html):



Further readings on this section:

- Chapter 6 of the textbook (**A Course in Phonetics** by Peter Ladefoged and K...
- Online sources:
  - http://clas.mq.edu.au/speech/phonetics/phonetics/airstream_laryngeal/...
  - http://savethevowels.org/posts/scuse_me.html

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CONSONANTAL GESTURES - I

At the end of this section, the students will be able to:

- EXPLAIN and DISTINGUISH among various types of articulatory gest production of English consonants.

Topic-104: Consonantal Gestures

In phonetics and phonology, speech sounds (segments) using basic gestures – they are treated as the abstract characterizations of articulatory dimension. Thus sounds (segments) are used to describe the phonological s and account for phonological variation. In this type of description in phonet the underlying units which are represented by classes of functionally e (gestures). Basically, the idea is to review the speech sounds involving thei purpose is to enhance the ability of the experts to describe unfamiliar sound appropriate description of unknown sounds is considered impossible) just i between varieties (of a language such as English) and among various langua involving the place of articulation and manners of articulation, sounds are exact locations and manners.

Topic-105: Articulatory Targets

A number of the possible places of articulation that are used in the been defined so far (in this course). The traditional terms which are used fo are not just names for particular locations (on the roof of the mouth). They s for articulatory targets. A large number of non-English sounds are to be four Iranian and African languages). Many of them involve using gestures in wh articulation is different from any similar sound found in English. For others manner of articulation that is different. Experts illustrate these different ty how each place of articulation is used in English and in other languages f fricatives (and treating them all as articulatory targets).

Topic-106: Types of Articulatory Gestures

Based on the definition of gesture as a possibility of the articulation contrast with English ones, the possible types of articulatory gestures are give

Bilabial gesture – very common in English (e.g., stops and nasal: p, b, m), Ewe of West Africa), bilabial fricatives contrast with labiodental fricatives. T and voiced bilabial fricatives are [ɸβ]. These sounds are pronounced by

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together, so that there is only a slit between them. Ewe also contrasts voiceless bilabia fricatives.

Labiodental fricatives – many languages (including English) have the labiodental fri probably no language has labiodental stops or nasals except as allophones of the corre sounds. In English, a labiodental nasal, [ɱ], may occur when /m/ occurs before /f/, *symphony*.

Dental sounds are present both in British and American English, e.g. dental fricatives no dental stops, nasals, or laterals except allophonically realized (before [θ, ð] as in *eig* Many speakers of French, Italian, and other languages (such as Urdu, Pashto and Sin dental stops such as [t̪ d̪]. However, there is a great deal of individual variation in th these consonants in all these languages.

Alveolar are very common targets and stops, nasals, and fricatives all occur in English languages at alveolar as a target of articulatory gestures (e.g., t, d, n, l, r., etc.).

Retroflex is very common sound in many Pakistani languages which is made by curli tongue up and back so that the tongue tip moves during the retroflex sounds such as [sounds are also the feature of Indian English.

Palato-alveolar and **palatal** are also possible articulatory gestures commonly found Similarly, velar sounds found in Urdu and other Pakistani languages need to be men [x, ɣ] which are velar fricatives. The gestures for pharyngeal (such as Arabic pharyn epiglottal sounds (such as epiglottal fricative [ʢ] involve pulling the root of the to back toward the back wall of the pharynx. Many people cannot make a stop gestur these sounds are found in Arabic and other Semitic languages.

Topic-107: Stops

Typical stop sounds are found in English and other languages of the wo other interesting types of stops which are found in languages other than English. N and see how rich are the stop sounds found in various languages:

Description	Symbol	Language
. Voiced	b	English and other languages
. Voiceless unaspirated	p	-do-
. Aspirated	pʰ	Sindhi and many other Pakistani
. Murmured (breathy)	bʱ	Sindhi
. Implosive	ɓ	Sindhi
. Laryngealized (creaky)	b̰	Hausa
. Ejective	k’	Hausa
. Nasal release	dn	Russian

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enasalized	nd	Swahili
ateral release	tɬ	Navajo
jective lateral release	tɬ’	Navajo
ffricate	ts	German
jective affricate	ts’	Navajo

e-108: Nasals

Nasal manners of articulation are commonly found in the languages of the world, can also occur voiced or voiceless (for example, in Burmese, Ukrainian and French sh and other most languages nasals are voiced. As voiceless nasals are comparatively ra olized simply by adding the voiceless diacritic [̥] under the symbol for the voiced soun cial symbols for voiceless nasals and it is written as /m̥/ - a combination of the letter fo l nasal and a diacritic indicating voicelessness.

109: Fricatives

Fricative as an articulatory gesture may be divided into voiced or voiceless sounds odivide fricatives in accordance with other aspects of the gestures that produce them. Fo uthorities have divided fricatives into sounds such as [s], in which the tongue is grooved n comes out through a narrow channel, and those such as [θ], in which the tongue wide slit through which the air flows. On the other hand, a slightly better way s is to separate them into groups on a purely auditory basis. Say the English voiceles ʃ]. Which two have the loudest high pitches? You should be able to hear that [s, ʃ] diff s way. The same kind of difference occurs between the voiced fricatives [z, ʒ] and [[s, z, ʒ, ʃ] are called sibilant sounds. They have more acoustic energy—that —at a higher pitch than the other fricative sounds.

readings on this section:

7 of the textbook (**A Course in Phonetics** by Peter Ladefoged and Keith Johnson) 5 and 7 of the additional reading book (**English Phonetics and Phonology-A Practica** Roach)

Mid Exam

End

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CONSONANTAL GESTURES-II

At the end of this section, the students will be able to:

- EXPLAIN and DISTINGUISH among various types of articulatory gestures production of English consonants.

Topic-110: Trills, Taps and Flaps

In approximants, trills, taps and flaps are also commonly found gestures in the world languages. These languages vary not only in terms of the as making it a usual /r/ approximant or unusual rhotic approximate [ɹ] found in also vary in terms of the length of the sound (some making it a short trill, other table covers various types of these approximant sounds found in the world lang

Voiced alveolar trill	r	Spanish
Voiced alveolar tap	ɾ	Spanish
Voiced retroflex flap	ɽ	Hausa
Voiced alveolar approximant	ɹ	English
Voiced alveolar fricative trill	ɹ̝	Czech
Voiced uvular trill	ʀ	French
Voiced uvular fricative or approximant	ʁ	Parisian French
Voiced bilabial trill	ʙ	Kele
Voiced labiodental flap	*	Margi

Topic-111: Laterals

As an important articulatory gesture, the central–lateral oppositio manners of articulation producing a lateral stop and a lateral fricative as which is by far the most common form of lateral sound. The only Englis British English, is /l/ with allophones [l] as in led [lɛd] and [ɬ] as in bell [bɛ English, initial [l] has more velarization than is typically heard in British E English, the air flows freely without audible friction, making this so approximant. It may be compared with the sound [ɹ] in red [ɹɛd], whicl alveolar central approximant. Laterals are usually presumed to be voiced statement to the contrary is made.

Topic-112: Summary of the Articulatory Gestures

A number of sounds share various features of speech production i represented by classes of functionally equivalent movement patterns

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ACOUSTIC PHONETICS-I

At the end of this section, the students will be able to:

- EXPLAIN and DEFINE acoustic phonetics, source/filter theory and tube models.

Topic-114: Acoustic Phonetics

As an important part of phonetics, acoustics is the study of the physics of the s hen speech sound travels through the air in a wave form) from the speaker's mouth rough vibrations. In acoustics, it is possible to measure and analyze these vi operties) by mathematical techniques usually by using specially-developed com oduce spectrograms of speech. Acoustic phonetics also studies the relationship betw eaker's vocal tract and the resulting sounds by involving physics, computer, statistic periments in lab. Thus, the analysis of speech using the expertise available in acc imed to be more objective and scientific than the traditional auditory method which reliability of the trained human ear. It also involves advanced level speech softv nd differences (in terms of pitch, loudness and quality) and distinguishes largely am giving the detailed composition of energy (e.g., frequency on spectrum).

pic-115: Explaining Acoustic Phonetics

Acoustics is a branch of phonetics which studies the physical properties o smitted between the mouth of a speaker and the ear of a listener. It is mainly devot hysics involving the latest technology such as computer and software (e.g., Praat) y of speech sounds wholly dependent on the use of instrumental (lab bas stigation. It particularly involves electronics and some grounding in physics; howe requisite for the advanced study of the subject. In this course, we are going to lea features of speech sounds (both consonants and vowels). Starting from the very cularly see the fundamental experiments of acoustics (such as recording of speech ately distinguishing among various forms and features of speech sounds. Thi duce you to the mostly used software in acoustic phonetics (i.e., Praat) and its etic analysis of human speech. Based on these introductory level experiments dur ted that the learners would be able to plan advanced level applications and experi

-116: Acoustic Analysis

According to the experts of speech sounds (phoneticians), acoustic analysis ive datum for the investigation of speech – the physical 'facts' of utterance. In ot nce is often referred to when one wants to support an analysis being made in arti tics. However, it is important to note down that one should not be too reliant c are subject to many mechanical limitations (e.g. the need to calibrate

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accurately), and which are often themselves open to multiple interpretations. Acou
gives us the features of a sound but also tells us about the duration or length of a sp
an analysis, we need to carefully know about material and procedure of recording. T
describes the durational characteristics, articulatory properties and phonetic
physiological measurement.

Topic-117: Acoustic Analysis of Vowels

The experts of phonetics are particularly interested in analyzing vowel
describe vowels in terms of numbers (in a language that how many vowels are
possible to analyze vowel sounds so that the measurement of the actual frequencies
formant structure of the vowels of a language) is taken. Having taken these formar
they can be represented graphically (plotting them on chart) as given for English vow
The figure (given in your textbook) gives the average of a number of authori
frequencies of the first three formants in eight American English vowels. The first tl
eight vowels of the American English words *heed, hid, head, had, hod, hawed, hoo*
taken. Explore the acoustic features of these vowels as recorded by the authors (Ladef
Subsequently (after your Praat sessions, at the end of this course), you will be able
vowels and try to see how your own vowels compare with these.

Topic-118: Source Filter Theory of Speech Production

Source-filter theory is an important concept in acoustic phonetics. It is a moc
owel) production. According to this theory, source refers to the waveform of the vi
ectrum is rich in harmonics, which gradually decrease in amplitude as their freque
rious resonance chambers of the vocal tract, especially the movements of the tongue a
yngeal source in the manner of a filter (see filtered speech), reinforcing certain har
ers. Thus the combination of these two elements (larynx as source and cavity as filte
urce-filter model of speech (e.g., vowel) production. We have already discussed that s
fer in pitch, loudness, and quality. Now if we understand the idea of source-filter we
lyze these changes as possible variation in speech sounds. When discussing differen
d that the quality of a vowel depends on its overtone structure (i.e., formants). Now
her way, we can say that a sound (e.g., vowel) contains a number of different pitche
e is the pitch at which it is actually spoken, and there are the various overtone pitch
nctive quality. We distinguish one vowel from another by the differences in these
ones are called formants, and the lowest three formants distinguish vowels from
of vowel production is explained in the next topic.

-119: Explaining Source – Filter Mechanism

In this theory, the tract is represented using a source-filter model and several de
l to synthesize speech in this way. The idea is that the air in the vocal tract acts li
ipe, or in a bottle. Sound travels from a noise-making source (i.e., the vocal fold v
en, at the lips, most of the sound energy radiates away from the lips for a listene

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ACOUSTIC PHONETICS-II

At the end of this section, the students will be able to:

- EXPLAIN and DEFINE perturbation theory and the acoustics of consonants.

Topic-120: Tube Models

Now in order to fully understand the concept of vowels and their formants, v
dea which is the tube model of vocal tract. Actually, the formants that characterize
ne result of the different shapes of the vocal tract. Any particle of air, such as that i
at in a bottle, will vibrate in a way that depends on its size and shape. Remember tha
act is set in vibration by the action of the vocal folds (in larynx). Every time the vc
ose, there is a pulse of acoustic energy (activation). Irrespective of the rate of vibrati
cal folds), the air in the vocal tract will resonate at these frequencies as long as the p
ans remains the same. Because of the complex shape of the filter (tract), the air w
n one way at once. So, the relationship between resonant frequencies and vocal trac
ch more complicated than the air in the back part of the vocal tract vibrating in one
r parts vibrating in another. Here we will just remember the fact that in most voi
ants are produced every time the vocal folds (source) vibrate. Note an interesting
nance in the vocal tract (filter) is independent of the rate of vibration of the vocal
words, the vocal folds may vibrate faster or slower, giving the sound a higher or lo
nts will be the same as long as the position of the tube (vocal tract) is the same. Th
ined in the next topic.

-121: Explaining the Tube Models

Actually this is an old idea (initiated by the great German scientist Hermann H
s merely the rapid repetition of its peculiar note. Now it is said that a vowel is the
onding to the vibrations of the vocal folds) of its peculiar two or three notes (fi
respectively). Actually, all voiced sounds are distinguishable from one another
(frequencies). This idea could be understood by considering the vocal tract as a tu
is when the vocal fold pulses have been produced at a steady rate, the "utte
. In other words, what you hear as the changes in pitch are actually the changes i
notone "voice." These overtone pitch variations convey a great deal of the qualit
he rhythm of the sentence is apparent because the overtone pitches occur only
ld have been vibrating. This tube model allows us to get an idea towards un
resonant frequencies (formants), the source-filter model and some other
between articulatory configurations (of vocal tract) and their acoustic consequer

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Keeping in mind this idea of perturbation theory, we can derive that the resonance frequencies will change (decrease or increase) as per the position (modification in the size and nature) of the vocal tract (tube).

Topic-124: Explaining Acoustic Analysis (Vowels)

Using computer programs, we can analyze vowel sounds by showing their components through the display (spectrogram). In spectrograms, time runs from left to right, the frequency of the components is shown on the vertical scale, and the intensity of each component is shown by the degree of darkness. It is thus a display that shows, roughly speaking, dark bands for concentrations of energy at particular frequencies—showing the source and filter characteristics of speech. Remember that the traditional articulatory descriptions of vowels are related to the formant frequencies. The first two frequencies are important here. The first formant (F1) is inversely related to the height of a vowel whereas the second formant (F2) is related to the frontness of a vowel sound. When the first two formants are taken, the vowels of a language can be plotted on a chart and the structure is very much related to the traditional description of vowel sounds.

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At the end of this section, the students will be able to:

- EXPLAIN and INTERPRET spectrograms of speech sounds a differences.

Topic-126: Explaining the Acoustics of Consonants

In addition to the above given features of consonant sou ideas are important to be remembered for interpreting the spectrog

Voiced - vertical striations corresponding to the vibrations of the v
Bilabial - locus of both second and third formants comparatively l
Alveolar - locus of second formant about 1700–1800 Hz.
Velar - usually high locus of the second formant
Retroflex - general lowering of the third and fourth formants
Stop - gap in pattern (with burst for voiceless and sharp formant b
Fricative - random noise pattern in higher frequency regions
Nasal - formant structure similar to that of vowels (with formants
Lateral - formant structure similar to that of vowels (with formant
Approximant - formant structure similar to that in vowels, usually

Topic-127: Interpreting Spectrograms

Words recorded in isolation are easy to be interpreted t spectrogram of connected speech is certainly a more difficult cha the students of phonetics and phonology learn spectrogram? spectrograms, the importance of learning and using spectrogram i

1. Using Praat (or any other software) and spectrogram is working on a problem related to the nature (physical pro or allophone?).
2. It increases our understanding of the speech sounds an isolation or as the part of connected speech).
3. Practice on spectrogram gives us the opportunity to le sounds.
4. It is also important for experts who are working on processing.
5. These are also used as the part of techniques in speech rec

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6. Spectrograms enable us to explore the complex nature of speech structure (
   language).
7. Spectrograms are the part of techniques used in machine translation.

In the light of the above points (and many more indeed), it is advisable to l
ated to spectrograms and increase your expertise as there are various resources av
pose.

ic-128: Useful Techniques for Interpreting Spectrograms

There are certain useful points to be remembered for interpreting segmen
h. We can start exploring them by looking at the segments one by one. In ad
oned above (in Topic-126), following are the helpful points to use in interpreting

Start analyzing sounds one by one by keeping in mind the individual charact
Topic-126) of sounds as a class.
Carefully see the overall structure, especially the frequency scale.
While interpreting consonants, also analyze the behavior of the adjacent vowels
Pay more attention to the first two formants (especially for vowels).
Watch for a burst and aspiration in stop sounds.
Remember that in vowels the first formant is inversely related to the height of a
s F1, the higher is the vowel) and F2 is related to the degree of backness of the
is, of course, also possible to tell many other things about the manner of arti
pectrograms of various sounds (e.g., one can usually see whether a stop has be
ricative, or even to an approximant in some cases). Similarly, the process of
op) can also be seen on many occasions. Trills can be separated from taps and
unds from voiceless ones.

: Individual Differences

garding the individual characteristics of sounds, we need to know some furth
times, we need to know about the idiosyncratic features of speech (when we a
pical speech community). We also need to discount, at times, the purely indivi
y significant speech. Moreover, for other particular reasons (such as us
ns), we need to discuss the validity of the identification of a particular spe
ow to study individual variation. Following points are important in this regard:

trograms show relative quality (e.g., a particular speaker may have a higher lev
formant plots (of an average speaker) may be compared with the formant plots
er.

arly, when two different speakers record their sets of vowels with the same ph
elative positions on a formant chart will be similar, but the absolute values o
ncies will differ.

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VOWELS AND VOWEL-LIKE ARTICULATIC

At the end of this section, the students will be able to:

- DEFINE and EXPLAIN vowels and vowel-like articulations with the help of pi cardinal vowel systems.

Topic-131: Vowels and Vowel-like Articulation

The fundamental distinction between consonant and vowel sounds is that ostruction to the flow of air. In addition to this, vowels are almost always fou llable, and it is very rare to find any sound, other than a vowel which can st llable. Phonetically, each vowel has a number of features (properties) that dist wels. These include; firstly, the shape of the lips (lip-rounding), rounded (for so utral (as for ə - schwa sound) or spread (as in /i:/ sound in word like sea or – ditionally ask you to say "cheese" /tʃi:z/ in order to make you look smiling. ' gue - the front, the middle or the back of the tongue may be raised, giving diffe npare /æ/ vowel (as in word 'cat') as a front vowel, with the /ɑ:/ vowel (as in 'c vel. Thirdly, the tongue (and the lower jaw) may be raised 'close' to the roof of vels. e.g. /i:/ or /u:/), or the tongue may be left 'low' in the mouth with the jaw or open vowels e.g., /a:/ and /æ/. In British phonetics, terms such as 'close' and els, whereas in American phonetics 'high' and 'low' are used for vowel descri three aspects are described in the case of vowels; lip-rounding, the part of the t e tongue. In addition to these three features, some other characteristics of vow us languages of the world (e.g., nasality – whether a vowel is nasal or not).

-132: Cardinal Vowels

In order to classify vowels (independent of the vowel system of a partic phonetician Daniel Jones introduced a system in early 20th century and wor called the "cardinal vowels" comprising of eight vowels to be used as refer wels could be related to them like the corners and sides of a map). Jones' idea a success and it is still used by experts and students for vowel description ed by the French phonetician Paul Passy, and it has been claimed that the set c imilar to the vowels of educated Parisian French of his time. Cardinal vowel d figure (the exact shape of which has been changed from time to time), wi en on the IPA chart from IPA website. It is a diagram to be used both for roun nd Jones proposed that there should be a primary and a secondary set of ca et includes eight vowels in total (from 1 to 8); the front unrounded vowels [i l vowel [ɑ] and the rounded back vowels [ɔ, o, u].

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Topic-133: Secondary Cardinal Vowels

Further on explaining vowels (particularly, the vow
secondary cardinal vowels (as a precise set of references) was
Danial Jones (1881-1967). Secondary cardinal vowels are
primary cardinal vowel system. The main difference betwee
related to lip-rounding as in some languages the feature of li
reversing the lip position (in comparison with primary cardi
types is produced (e.g., rounding the lips for the front vowels)
(from vowel 1 to 8), following are the secondary cardinal vo
pointed out Danial Jones:

9. Close (high) front rounded vowel	[y]
10. Close-mid front rounded vowel	[ø]
11. Open-mid front rounded vowel	[œ]
12. Open (low) front rounded vowel	[æ]
13. Open (low) back rounded vowel	[ɒ]
14. Open-mid back unrounded vowel	[ʌ]
15. Close-mid back unrounded vowel	[ɤ]
16. Close (high) back unrounded vowel	[ɯ]

There are two more vowels representing the highest po
These are secondary cardinal vowels with numerical codes 17 a
respectively.

Topic-134: Comparing Primary VS Secondary Cardinal Vo

As discussed in Topic 133, the secondary cardinal vo
with primary cardinal vowel system. Remember that secondary
ɒ] are the rounded counterparts of primary cardinal vowels fro
cardinal vowels from 14 to 16 (ʌ, ɤ, ɯ) are the unrounded equi
to 8 [ɔ, o, u] respectively. Two further cardinal vowels (17 and 1
highest point at the center where the tongue can possibly rea
languages) is usually shown in the form of the cardinal vowel
divided into eight corners) or cardinal vowel quadrilateral.
configuration of the degree and direction of tongue movement
diagrams have been successfully used for the description of v
languages of the world.

Further readings on this section:

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While discussing vowels in English, we can explore [illegible] of reasons. The first and the foremost comparison may be with the American English (or varieties of American English). Both of the varieties have been discussed by linguists and politicians alike (from both sides). The vowels of BBC accent are different in both number (20 in total – both pure vowels and diphthongs) and quality. British English speakers distinguish the vowel [ʌ] in cut from the vowel [ɜ] in curt. It does not have any r-coloring (rhotacization) mainly by the frequency of the first formant. Moreover, as the main feature of BBC English (to be noted here) is the distinction between the three back vowels [a] as in father and cart, [ɑ] as in bother and cot, and [ɔ] as in author and caught. BBC vowels are really interesting in making it a standard variety of English.

Topic-137: Vowels in Other Languages

Vowels in languages other than English are important to be explored here in connection with the on-going discussion. Variations among human languages are not limited only to the number of vowels but also to the quality and features. The vowel systems of some of the major languages are briefly mentioned here. Spanish has a very simple system contrasting only five vowel sounds [i, e, a, o, u]. It is important to

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remember that these symbols do not have the same values as English or their descriptions of cardinal vowels. Japanese also has a set of five vowels [i, e, a, o, u] but very different in a narrower transcription than that of Spanish. Similarly, Danish has different vowels and their qualities as it contrasts three front rounded vowels which may occur both in long or short forms. Focusing on Pakistani regional languages, Urdu and Punjabi have very different nasal vowels (the nasality is phonemic here). Urdu has some seven nasal vowels such as /he/ vs. /hẽ/ or /hi:/ vs. /hĩ:/ as in word /nahĩ:/ meaning 'no'. Nasalization in vowels is a common feature of Indo Aryan languages.

Topic-138: Advanced Tongue Root (ATR)

Normally, while discussing the degree of variation in vowel sounds, three types of features are given (i.e., height of tongue, backness of tongue and lip rounding) which cover the major variation in world's languages. But this description does not cover all types of variation in vowel quality. One out of such variations is advanced tongue root (ATR) which is found in Akan language spoken in Ghana. with ATR involve the furthest-back part of the tongue, opposite to the lved in the production of speech sounds - also called the radix ATR (a kind of articulation in which is used

VOWELS AND VOWEL-LIKE ARTICU

At the end of this section, the students will be able to:

- DEFINE and EXPLAIN nasalization, semivowels and secondary arti vowels.

Topic-140: Nasalization in Vowels

In the description of vowels, nasalization is another feature whicl method of description as it is not found in all languages of the world. Th many other Pakistani regional languages learn to produce a variety of r mother tongue and face no issue in learning nasalization in vowels. Ho languages (such as English which does not have nasal vowels) have to starting saying the low vowel /æ/ as in man and by keeping the soft palate contrasts between nasal and oral vowels including French and Urdu. Urd vowels. Urdu has seven nasal vowels such as /he/ (meaning 'is') vs. /hẽ/ vowels is a common feature of Indo Aryan languages. In IPA chart, the the symbol [˜] called tilde (used above the phonetic symbol to show the na

Topic-141: Summary of Vowel Quality

In this course, we have discussed many features related to the descr topic, we provide a summary of all these features related to vowels. Th quality (i.e., height and backness of the tongue) that are used to contrast or all languages of the world. But there are four other features that are us languages exhibit them. They include 'lip-rounding', rhotacization, nasaliz (ATR). The following are the acoustic correlates of all these six features:

- Height - frequency of formant one (inversely related to F1)
- Backness - difference between frequencies of F2 and F1
- Rhotacization - frequency of formant three
- Rounding - lip position (rounded, half rounded or neutral)
- ATR - width of the pharynx (ATR or RTR)
- Nasalization - position of the soft palate

pic-142: Semivowels

Most of the world languages contain a class of sounds that fu nsonants but is phonetically similar to vowels (e.g., in English, /w/ and /j/ y are used in the first part of syllables (at onset), they function as

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pronounced slowly, they resemble (in quality) with the vowels [u] and [i]
called semivowels which are also termed as approximants today. In Fren
i.e., in addition to j and w there is another sound symbolized /ɥ/ and is
word like 'huit' /ɥit/ (eight) and in consonant clusters such as /frɥ/ in /frɥ
ists a semivowel corresponding to the back close unrounded vowel /ɯ/. Li
n approximant.

Topic-143: Secondary Articulatory Gestures (SAG)

While discussing vowels, we need to highlight some of the se
owels. Remember that 'secondary' articulation is an articulatory gesture
ccurring at approximately the same time as another (primary) gesture. It
which is at the same time and of the same value (taking place as an e
ppropriate to consider four types of secondary articulations in conjunctio
sually be described as added vowel-like articulations including; 'palat
ongue gesture as in sound /i/), velarization (raising of the back of the tong
uperimposition of the narrowing of the larynx) and labialization (the addit

Topic-144: SAGs Discussion

The four types of possible secondary articulatory gestures related
here for understanding and further detailed exploration. Palatalization i
ongue gesture (as in sound like [i]) to another main (primary) ge
palatalization is the small [ʲ] superimposed above another symbol (for
palatalization (a process whereby the place of an articulation is shifted n
palate) and palatalized (when the front of the tongue is raised close to t
closure is made at another point in the vocal tract) are sometimes used
palatalized consonant has a typical /j/-like (similar to /i/ vowel) quality.

Velarization involves raising the back of the tongue (adding the /u
considered as the addition of an [u]-like tongue position (but remember t
the lip rounding). A typical English example of velarization is the /l/ soun
words like kill, pill, sell and will) called velarized or dark /l/ and may be
velarization are both [ˠ] and [~].

Another secondary gesture of vowel quality is pharyngealization
narrowing of the pharynx. The IPA diacritics for symbolizing phar
velarization) and [ˤ] (the superimposition of the symbol for pharyngeal sou

The last secondary articulatory gesture for vowel quality to be note
the addition of lip rounding (written as [ʷ]) to other primary articulatio
Remember that nearly all kinds of consonants can have added lip roundin
have one of the other secondary articulations (such as velarization and pala

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Topic-145: Summary of the SAGs

Types of SAGs:-

The secondary gestures for vowel quality are hereby summarized fo
may or may not have any of the four secondary articulations su
pharyngealization and labialization. Labialization may also have (at th
secondary articulatory gestures (and even if the sound is itself labial such
these secondary articulatory gestures for vowel sounds are briefly mention

1. Palatalization [ʲ] is the raising of the front of the tongue such as fo
2. Velarization (written as [ˠ] and [˞]) is the raising of the back of the
3. Pharyngealization [ˤ] is the retracting of the root of the tongue.
4. Labialization [ʷ] is the rounding of the lips such as Arabic [sʷ] an

Further readings on this section:

- Chapter 9 of the textbook (**A Course in Phonetics** by Peter Ladefoged a
- Chapters 3 and 14 of the additional reading book (**English Phonetics an Course** by Peter Roach)

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SUPRASEGMENTAL FEATURE

At the end of this section, the students will be able to:

- DEFINE, EXAMINE and EVALUATE suprasegmental features such

Topic-146: Introduction to Suprasegmental (SS) Features

Supra means 'above' or 'beyond' and segments are sounds (phon used in phonetics and phonology to refer to a vocal effect (such as tone extends over more than one sound (segment) in an utterance. Major s pitch, stress, tone, intonation or juncture. Remember that these features applied above segmental level (on more than one segment). Phonological fields: segmental phonology and suprasegmental phonology. Supras extensively explored in the recent decades and many theories have l application and description of these features.

Topic-147: Syllable

The 'syllable' is a fundamentally important unit both in the fiel Experts normally keep the phonetic notions of the syllable separate from to understand but very difficult to define the syllable. In a simple way of the parts of word (in which a word is further divided into parts), for exan *seg-men-tal*. Phonetically, we can observe that the flow of speech typi between vowel-like states (where the vocal tract is comparatively open a like states where some obstruction to the airflow is made (thus altering kinds of sounds). So, from the speech production point of view, a syllabl constricted or silent state to a vowel-like state and then back to cons acoustic point of view, this means that the speech signal shows a series o to vowel-like states separated by troughs of lower energy (sonority).

Topic-148: Explaining Syllable

Phonologists are interested in the structure of a syllable. It can be as phonemes may occur at the beginning (onset), in the middle (nucleus syllables - the combination of nucleus (peak) and coda is called the rh ending (coda) are optional while a syllable must have a nucleus (at least the sequences of phonemes is called phonotactics, and it seems that t language are determined by its syllabic structure (sequences of sounds th be broken down into syllables). Syllables are claimed to be the most basi a sound system) meaning that an utterance must contain at least one sy

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has syllables, and children learn to produce syllables before they can manage t language. It is important to note that syllable structure could be of three types: (CVC) and 'complex' (with consonant clusters at edges) such as CCVCC and vowel and C stands for consonant). Moreover, words can have one syllable (m (bisyllabic or disyllabic), three syllables (trisyllabic) or many syllables (polysyl

Topic-149: Stress as a Suprasegmental Feature

Stress is basically a suprasegmental feature applied to a whole syllable by adding factors such as loudness, rise in pitch, length of duration and vowe other syllables). For example, in mi-ni-mi-**ZA**-tion, the second last (penultimate greater amount of energy is applied to it) as it is louder and longer than the r word. Similarly, its quality and pitch are different than that of others. All thes (compare: **IN**-sult vs. in**SULT** and be-**LOW** vs. **BILL**-ow). Stress is an i phonetics and phonology. Despite the fact that it has been extensively studied, tl disagreement among the experts. To begin with a basic point, it is almost languages some syllables are in some sense stronger than other syllables; these potential to be described as stressed. Stress plays an important role in conveying

Topic-150: Types and Categories of Stress

One of the areas in which a little agreement is found is related to th descriptions of languages manage with just two levels (stressed and unstressed than two. In English, one can argue that if one takes the word 'in-di-ca-tor' syllable is the most strongly (primarily) stressed one, the third syllable is (secondarily) stressed and the second and fourth syllables are weakly stressed or This gives us three levels (primary, secondary and tertiary) and it is possible to this rarely seems to give any practical benefit. Remember that the primary str languages like English as it has the ability to change meaning e.g., in two-syllal nsult, meanings and the category of the words are changed depending upon the p o the syllables: INsult (noun) vs. inSULT (verb); whereas in long words (suc yllable receives primary stress while other (one or more than one syllable) h econdary stress is just for better pronunciation (not changing the meaning). A etween lexical stress (phonemic in nature) and sentence level (emphatic) stress. ext topic.

opic-151: Lexical and Emphatic Stress

In terms of its linguistic function, stress is often treated under two di xical) stress and sentence (emphatic) stress. Lexical stress is basically relate plied at syllable level (when only one syllable is stressed) that has the ability to c grammatical category of a word as in the case of 'IMport' (noun) and 'imP el stress, on the other hand, is applied on one word (rather than a syllable) in a word more prominent (stressed) than the rest of the words in the sentence. Thi

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SUPRASEGMENTAL FEATURES-

At the end of this section, the students will be able to:

- Define syllable-timed and stress-timed languages.

Topic-152: Stress Timed Languages

It is sometimes claimed by the experts that different languages (and of rhythmic patterns. Languages of the world are; therefore, divided into timed language and syllable timed languages. But this is basically the divisi of their modes of timing (i.e., stress vs. syllable timed languages). Stress tim their dominating rhythmic feature meaning that these languages seem to stressed patterns (the division among the syllables is made on the basis of stre e.g., English and German languages). In other words, in stress timed languag with regular intervals and their units of timing are perceived accordingly. St these rhythmical types, and is said to be characterized by a tendency for st equal intervals of time. This idea is further clarified in the next topic.

Topic-153: Explaining Stress Timed Languages

'Stress timed languages' is a very general phrase used in phor pronunciation of languages displaying a particular type of rhythmic pattern syllable-timed languages. In stress-timed languages, it is claimed that the regular intervals of time (stress-timing) regardless of the number of intervening English. This characteristic is sometimes also referred to as 'isochronism', or clear that this regularity is the case only under certain conditions, and the ext towards regularity in English is similar to that in, say, other Germanic lang short, the division among the syllables is made on the basis of stress and un anguages, stress is realized both at word and sentence levels approximatel patterns (particularly at sentence level).

Topic-154: Syllable Timed Languages

In syllable timed languages, all syllables tend to have an equal time ngth or duration) and the rhythm of the language is said to be syllable-tir llables tend to occur at regular intervals of time with fixed word stress. A class ich all morae have approximately the same duration. This tendency is cont ere the time between stressed syllables is said to tend to be equal irresp tressed syllables in between. Czech, Polish, Swahili and Romance langu

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are often claimed to be syllable-timed. Many phoneticians; however, doubt wh
yllable-timed.

5: Explaining Syllable Timed Languages

yllable timed is a very general term used in phonetics to characterize the
; displaying a particular type of rhythmic pattern. It is opposed to stress-tir
imed languages, the syllables are said to occur at regular intervals of time, (e.
stic is sometimes referred to as isosyllabism or isosyllabicity. However, ve
: on the accuracy or general applicability of such properties, and the usefulne
questioned. So it is important to remember that the division of languages in tw
s. stress timed languages) is not based on strong concepts. Many phoneticians
of timing value. They are of the view that there are three dimensions: fixed w
Romance languages), variable word stress (mainly found in languages such
and fixed phrase stress (phrase as a third possibility as exhibited by Japanese)
languages on the basis of these three patterns. write 3 dimensions

6: Pitch as a Suprasegmental Feature

s a suprasegmental feature, pitch is an auditory sensation - when we hear a re
h as a note played on a musical instrument (or a vowel produced by the human
ı (when the rate of vibration is high) and a low pitch (when the rate of vibratic
speech sounds that are voiceless (e.g. /s/), and cannot give rise to a sensation of
oiced sounds can. Thus the pitch sensation that we receive from a voiced so
ely to the frequency of vibration of the vocal folds. However, we usually refe
as fundamental frequency in order to keep the two things distinct. In tonal la
n essential component of the pronunciation of a word and a change of pitch ma
ng. In most languages (whether or not they are tone languages) pitch plays
ı. In very simple words, pitch is the variation in the vibration of vocal folds.

readings on this section:

10 of the textb ok (**A Course in Phonetics** by Peter Ladefoged and Keith John
s 15 to 19 of th additional reading book (**English Phonetics and Phonology-A
by Peter Roach)
ources: http://sl deplayer.com/slide/3885548/
https://v ww.slideshare.net/shassan573/pronunciation-39434371
http://ec concord.ied.edu.hk/phonetics_and_phonology/wordpress/?page

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SUPRASEGMENTAL FEATURES-III

At the end of this section, the students will be able to:

- DEFINE, EXAMINE and EVALUATE suprasegmental features such as into

Topic-157: Tone and Tonal Languages

Tone (in phonetics and phonology) as a suprasegmental feature re movement (variation) or level of pitch that is used in a linguistically contrasti languages, the linguistic function of tone is to change the meaning of a word. Fo Chinese, [ma] said with a high pitch means 'mother' while [ma] said on a low ris In other (non-tonal) languages, tone forms the central part of intonation, and the example, a rising and a falling tone on a particular word may cause a differe entence in which it occurs. In the case of tone languages, it is usual to identify to f individual syllables, whereas an intonational tone may be spread over many syll nglish intonation, tone refers to one of the pitch possibilities for the tonic (or rther analysis, a set of four types of tone is usually used (fall, rise, fall–rise and e also suggested by various experts. Four possible types of tone?

pic-158: Intonation as a Suprasegmental Feature

In simple sense, 'intonation' refers to the variations in the pitch of a sp vey or alter meaning (at sentence level). In its broader and more popular sens h the same field as 'prosody', where various features such as voice quality, te included. It is a term frequently used in the study of suprasegmental phono nctive use of patterns of pitch, or melody and the study of intonation is sometin rts have suggested several ways of analyzing intonation. In some approaches, ibed as contours and analyzed in terms of levels of pitch as pitch phonemes and s, the patterns are described as tone units or tone groups which are further ana ar tone, tonicity, etc. The three variables of pitch range, height and dir guished. Some approaches, especially within pragmatics, operate with a much f the tone unit: intonational phrasing is a structured hierarchy of the intonati sation. A formal category of intonational phrase is also sometimes recog ce span dominated by boundary tones.

159: Functions of Intonations

ation according to grammatical structure?

Intonation as a suprasegmental feature performs several functions in a l nt function is to act as a signal of grammatical structure (e.g., creating patt rammatical categories), where it performs a role similar to punctuation (in w

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more contrasts (for conveying meaning). Intonation also gives an idea about the syntacti
tence, clause and phrase level boundaries). It also provides the contrast between som
uctures (such as questions and statements). For example, the change in meaning illustrate
king me or telling me?' is regularly signaled by a contrast between rising and falling pitc
of intonation in sentences like 'He's going, isn't he?' (= I'm asking you) opposed to 'He
!' (= I'm telling you) (These examples are given by Peter Roach). The role of intonation
cation is quite important as it also conveys personal attitude (e.g., sarcasm, puzzleme
inally, it can signal contrasts in pitch along with other prosodic and paralinguistic featu
ing variation in meaning and can prove an important signal of the social background of

Explaining Functions of Intonation

ile discussing the functions of intonation, another approach is to concentrate on its r
nal discourse. This involves such aspects as indicating whether the particular thing bein
new information or old (in sentence, for example). It further creates the regulation o
conversation, the establishment of dominance and the elicitation of co-operative respo
with the signaling of attitudes, it seems that though analysts concentrate on pitch mov
many other prosodic factors being used to create these effects. It is also important to n
a less work has been done on the intonation of languages other than English. It seems
s have something that can be identified as intonation (there can be many differences
anguages). It is, therefore, a potential area to be studied and explored further. One re
on not being well explored is because of the different descriptive frameworks used by
for studying intonation cross-linguistically. Even it is claimed that tone languages
on and that it is superimposed upon the tones themselves. Such a claim creates especiall
ns of analysis.

er readings on this section:

pter 10 of the textbook (**A Course in Phonetics** by Peter Ladefoged and Keith Johnson)
pters 18 and 19 of the additional reading book (**English Phonetics and Phonology-
actical Course** by Peter Roach)

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LINGUISTIC PHONETICS-I

At the end of this section, the students will be able to:

- DEFINE and EXPLAIN linguistic phonetics in the broader perspective of IPA.

Topic-161: Linguistic Phonetics

Linguistic phonetics is an approach which is embodied in the principles Phonetic Association (IPA) and in a hierarchical phonetic descriptive framework basis for formal phonological theory. Speech, being very complex phenomena and h of organization, needs to be explored from different angles. Linguistic phonetics a questions related to the possible ways of articulatory unified phonetics and phon perspective of cognitive phonetics focusing on speech production and perception anguages as a sound systems. The idea is mainly related to the overall ability of hum ounds (as a community and irrespective of their specific languages) and then the re nared knowledge (as considered by the IPA in its charts) for formal phonetic and pho

pic-162: Phonetics of the Community and of the Individual

Linguistic phonetic descriptions (of speech sounds) are, by and large, onetics of the community (excluding the individual properties and considering the sound system of a language by its native speakers). The representations that expe IPA, and analyze in a formal phonological theory, are intended to show the wledge of how to say the words of a language. It is important to note that t wledge is perceptible to other speakers (and thus to the phonetician as well) and ed to the aggregate behavior of linguistic group in the sense that it capture bers accept as correct pronunciation system. So the focus of linguistic phoneti etics of the community and the phonetics of the individual is considered only fo n required).

c-163: Explaining Phonetics of the Community and of the Individual

f: Phonetic of Community is considered for Phonetic

The major reason why the phonetics of the community is considered for phor irstly, individual speakers differ in interesting ways (two native speakers of a la with some variations). The description of the phonetics of the individual invo tic knowledge and skills related to the performance of language. It is possible tha onetics of the individual can be captured using IPA transcription but others are n as his private knowledge and its performance and the role of memory and exp onetics of the individual is usually not the focus of the linguist in speech el t to describe even with spectrograms of the person's speech. Although, the

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individual is the focus of much of the explanatory power of phonetic t
description we need to focus on the phonetics of the community.

Topic-164: The IPA

Fill in blanks

While discussing the key elements of linguistic phonetic descri
International Phonetic Alphabet (abbreviated as IPA). IPA is the set of s
been officially approved by IPA. The association publishes a chart comp
charts. At the top inside the front cover, you will find the main conso
showing the symbols for nonpulmonic consonants, and below that is th
cover is a list of diacritics and other symbols, and a set of symbols for s
such as tone, intonation, stress, and length. Remember that the IPA chart d
types of phonetic descriptions (e.g., all the individual strategies for re
contrasts, or gradations in the degree of co-articulation between adjacer
limited to those possible sounds that can have linguistic significance in th
of a word in some languages. So the description of IPA is based on
community.

Topic-165: Explaining the IPA

Fill in blanks

One evidence that the IPA chart is based on linguistic phonetics
cells on the chart (those neither shaded nor containing a symbol) tha
categories that are humanly possible but have not been observed so far to
(e.g., a voiceless retroflex lateral fricative is possible but has not been
blank). The shaded cells, on the other hand, exhibit the sounds not possibl
the consonant chart is a set of symbols for consonants made with differen
voiced implosives, and ejectives). All these descriptions reflect the potent
(as a linguistic community) not only showing the possible segments but a
and points related to the possible airstream mechanisms and even the
articulations and secondary articulatory gestures. The IPA chart is careful
is continuously revised and updated.

Further readings on this section:

- Chapter 11 of the textbook (**A Course in Phonetics** by Peter Ladefoged

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LINGUISTIC PHONETICS-II

At the end of this section, the students will be able to:

- DEFINE and COMMENT on 'feature hierarchy' and problems with linguistic exp

Topic-166: Feature Hierarchy

The second way by which we will consider why the linguistic phonetic descr by considering a hierarchical organization of linguistic phonetic features. Feat important concept in phonetics and phonology which is based on the properties and a very general sense, a feature may be tied to a particular articulatory maneuver or a example, the feature [bilabial] indicates not only that the segment is produced wit involves both of them. Such features (in phonetics and phonology) are listed in a hi the hierarchy defining ever more specific phonetic properties. For example, sounds their supra-laryngeal and laryngeal characteristics, and their airstream mechanism characteristics can be further divided into those for place (of articulation), manne possibility of nasality, and the possibility of being lateral. Thus, these features a speech sounds and describing them formally. Explain with reference to into supra laryngeal n laryngeal?

Topic-167: Feature Hierarchy: Discussion

major features based on major region of vocal tract?
For dividing speech sounds through feature hierarchy, the first division the major regions of the vocal tract, giving us the five features in total (i.e., L Radical, and Glottal). The first three of these features are related to tongue positio cover term for [pharyngeal] and [epiglottal] articulations made with the root of th 'Glottal', on the other hand, is based on being [glottal], to cover various articula are to have a convenient grouping of the features for consonants, we have t Laryngeal features must allow for the dual nature of the actions of the larynx place of articulation. Remember that a sound may be articulated at more than o Coronal, Dorsal, Radical, and Glottal. Within the five general regions, 'Coronal' into three mutually exclusive possibilities: Laminal (i.e., blade of the tongue) tongue), and Sub-apical (i.e., the under part of the blade of the tongue). Thus t subdivided into sub regions on the basis of their features. Coronal three

Topic-168: Feature Hierarchy: Explanation

This grouping under feature hierarchy also reflects the fact that the Fricative, Approximant, and Vowel) depend on the degree of closure of the art grasp the feature hierarchy, we need to explore the major divisions in this to 'Stop' has only one possible value (i.e., [stop]), but 'Fricative' has two (i.e., [s

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...ardy, 'Approximant' and 'Vowel' have five principal features: Heig
...gh], [mid-high], [mid], [mid-low], and [low]), Backness (with three
...k]) and two kinds of 'Rounding' (i.e., Protrusion with possible values [p
...mpression, with possible values [compressed] and [separated]). The fe
...proximants is the 'Tongue Root' which has two possible values: [+AT
...ature 'Rhotic' has only one possible value, [rhotacized]. It is also imp
...aryngeal' possibilities involve mainly five features ([voiceless], [breathy v
...ice], and [closed] - forming a glottal stop). Airstream features have three v
...glottalic).

Topic-169: Problems with Linguistic Explanations

While discussing the phonetics of the individual, we need to conside
explanation does not always satisfy all the topics of speech production and v
the phonetics of the individual. There are current phonetic research trends an
large extent on topics such as speech motor control, the representation of
interaction of speech perception and production and their roles in language ch
find explanations for language sound patterns in these and other related topi
phenomenon of 'assimilation' where the adjacent sounds come to share some
we restrict ourselves to the terminology and knowledge base of linguistic phc
community); we are restricted to descriptions of sound patterns and not
explanations that are restricted in this way often fall into the fallacy of reificati
things are concrete).

Topic-170: Explaining the Problem with Linguistic Explanations

The phonetics of the individual gives explanation to a number of
example, take the case of 'assimilation' - it happens because there is a ten
adjacent sounds to share phonetic properties (at individual level). This "
impressive if we state it as a formal constraint (in Optimality theory) on s
following:

AGREE(x): Adjacent output segments have the same value of the feature x.

It means that assimilation exists when adjacent segments share featu
say that the adjacent sounds would influence the sounds in the neighborho
assimilate (a cross-linguistic generalization) exists because there is a tendenc
specific "explanatory principle"). The private phonetic knowledge of the in
satisfying way to explain language sound patterns.

Further readings on this section:

Chapter 11 of the textbook (**A Course in Phonetics** by Peter Ladefoged and

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experiences are encoded in memory. As per the phonetic implement
memory in their most basic phonetic form and used when needed.

Topic-173: Explaining Memory for Speech

The role of the memory for speech under the exemplar the
each word are stored in memory and their phonetic variability is m
main postulates of the concepts are given here:

- Language universal features: Broad phonetic classes (e.g., as
  physiological constraints on speaking or hearing, but thei
  arbitrary—a matter of community norms.

- Speaking styles: No one style is basic (from which others are
  memory. Bilingual speakers store two systems.

- Generalization and productivity: Exemplar theory says that ge
  productivity. Interestingly, productivity—the hallmark of ling
  implementation approach—is the least developed aspect of the

- Sound change: Sound change is phonetically gradual and operat
  gradual shift as new instances keep on adding.

Topic-174: The Balance Between Phonetic Forces

In order to explain the sound patterns of a language, the views
considered. Both of them like to use the least possible articulatory effor
produce very clear speech), and there are a large number of assimilatio
and other reduced to minimum. Thus a speaker uses language with a
articulation and secondary articulation). This tendency to use language
ease of articulation leads to change in the pronunciation of words. In
change in the place of the nasal and the following stop occurred in
impossible before these words came into English through Norman Frenc
that occurs in the prefix in- (as in intolerable and indecent) has changed
reflected in the spelling. In all this and in many similar historical chan
affected by adjacent segments so that there is an economy of articulation.
phenomenon of assimilation. Similarly, articulation with the least possib
keeping the balance between phonetic forces ultimately leading to ch
sounds. This idea is explained in the next topic.

Topic-175: Explaining the Balance Between Phonetic Forces: Explanat

While going for producing sounds with maximum ease of articul
affected. The focus of the speakers is always on maintaining a sufficient p

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At the end of this section, the students will be able to:

DEFINE and EXPLAIN phonotactics and syllabic templates.

Topic-176: Syllabic Templates and Syllabification

Syllable is the unit of pronunciation typically larger than a sin word maybe divided into parts such as in *ne-ver-the-less*, and a g ese syllabic divisions occur in writing, thus providing informat yphenated (separated into syllables). Syllabification, on the other ha ivision of a word into syllables. Similarly, resyllabification refers ication of syllable boundaries. A word containing a single syllable i ontains more than one, the term polysyllabic is used. Remember onsonant is found at the onset followed by a vowel as its peak) of s ie world. It is the universal pattern of syllable (Max Onset C) and is e n abundance. There are languages which only allow CV templa VCVCVCV and Waikiki - CVCVCV). Interestingly, it is also found anguages of the world: *kami, nana, baba, papa, mani, rani, etc.* hildren first acquire the CV pattern of their mother tongue.

What specific terms

Topic-177: Explaining Syllabification

Clusters Cc and C

When it comes to the internal structure of syllable, langua syllabic patterns are different across languages and language famili structures for syllable structure and languages are labelled as per t sequences are called clusters (e.g., CC – two consonants or CCC phonotactic analyses are based on the syllable structures and syllab consonant clusters, mainly three types of syllabic patterns are cons moderate – complex (on the basis of consonants clusters at edges: c examples:

Simple	CV
Moderate	CVC(G)(N) (G for Glide and N for Nasal –
Complex	CCVCC – CCCVCC (bipartite CC and tripa

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Topic-178: Phonotactics

The study of the phonemes and their order found in the syllables (the
of a language is called the phonotactics. It has often been found that languages
to appear in any order (e.g., a native speaker of English can figure out fairly
phonemes /streŋθs/ makes an English word ('strengths') and that the sec
acceptable as an English word 'blage', although that word does not happen
lvm/ could not possibly be the part of an English word). Phonotactic analyses
ome interesting findings. For example, why should 'bump', 'lump', 'hump', 'r
nd others all be associated with large blunt shapes? Why should there be a wh
ith a plosive and a syllabic /l/ all having meanings to do with clumsy, awkwa
nuddle', 'fumble', 'straddle', 'cuddle', 'fiddle', 'buckle', 'struggle', 'wrig
llables begin with /pw/, /bw/, /tl/, /dl/ when /pl/, /bl/, /tw/, /dw/ are acceptat
lled the phonotactics of the language.

9. Explaining Phonotactics

Phonotactics is a term used in phonology to refer to the order (sequen
avior) of segments (sounds or phonological units) which occur in a languag
phonologically well-formed structure of a word. The allowed sound pat
rns of language are found through phonotactics. For example, in English,
s/ and /spm/ do not occur initially in an English word, and there are man
ble consonant+vowel combinations which may occur. By thoroughly
ential constraints' of a language can be stated in terms of phonotactic
rative phonotactics, no phonological principles can refer to morp
logical patterns which are sensitive to morphology (e.g. affixation, etc.) a
ological component of the grammar (not in the phonology). Some ex
tactics are given for your understanding:

oneme pattern	V	I, oh, owe
oneme pattern	CV	to, be, see
honeme pattern	CVC	cat, dog, run
oneme pattern	CCVC	stick, click, brick
neme pattern	CCVCC	brisk, treats, speaks
eme pattern	CCCVCC	streets, strand, strips
oneme pattern	CCCVCCC	strengths

sible: CCV (try) CCCVC (stroke), CCCV (straw), VCC (eggs) CVCC (

readings on this section:

urces: http://people.umass.edu/scable/LING201-SP13/Slides-Handouts/
Phonotactics.pdf
http://clas.mq.edu.au/speech/phonetics/phonology/syllable/syll_p
https://en.wikipedia.org/wiki/Syllabification

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ing downloaded Praat, now start going through the two major parts
lows) and explore various components of Praat.
to spend time and explore further. Now do the following:

- Open Praat: double click on the icon.
- Get to know the Praat layout (explore the buttons on the object windo
  and synthesis tools and the options on the picture window).

ic-183: Recording and Displaying Long / Short / blanks

This is now your first practical on Praat. We are going to learn how to r
display the recorded file in this experiment. We will also know about the g
work on the object window and sound window and use the edit button
the tools panel).

- Go to NEW > RECORD mono-sound (with sampling rate – 44100 Hz)
- Make sure the volume bar is fluctuating as you record – if it isn't, y
  don't see the volume bar at all, you're not speaking loudly enough.
- Watch out for clipping. If your recording level is too high and you g
  bar, you'll end up with what is called a "clipped" signal; this is very b
- Give the recording a name (in the box below "Save to list").
- Save to list.

gratulation! You have recorded your first sound file. Now in order to disp

- Open the sound file in the Edit window.
- In the Objects window, highlight the sound file you've just recorded
- Click Edit on the "Analysis and synthesis tools" panel.

final step of your Lab 1, you're now required to EXTRACT from this f

- Select the portion that you want to study.
- File > Extract Selection (The extracted selection will show up as an
  Praat objects window).
- Close Edit window.

have completed all the steps of the experiment I. While conducting the
recordings on these topics.

pic-184: Segmenting and Labeling

'Segmenting and labelling' is our next experiment on Praat w
ustic analysis because we will be able to place segmental symbols
ctrogram) and annotate the sound file. The sound file needs text for

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USING PRAAT-II

...he end of this section, the students will be able to:

EXPLAIN the Source-Filter model of speech
RECOGNIZE and ANALYSE features such as fundamental frequency, harm

...opic-186: The Source-Filter Model of Speech

We have been theoretically discussing about the source-filter theory in t ...uched upon the theoretical aspects such as 'the tube model' (of the vocal tra ...eory' on the formant structure of vowels. Source-filter theory is particularly im ...asic components of speech sounds and the nature of the acoustic signals (it ...ounds). It is, therefore, very crucial to understand the acoustic analysis of speec ...owels and vowel-like (sonorous sounds). The major goal of this lab is to un ...asic acoustic components of (sonorous) speech sounds such as fundamental fr ...ormants. Basically, these components create the acoustic signals associated wit ...hem is crucial to understanding what we actually hear, when we hear speech s ...he source-filter theory are conducted in the next few sessions.

Topic-187: Measuring the Fundamental Frequency

Fundamental frequency (F0 or Pitch) is an important component of sou ...aken from the middle of the sound (e.g., vowel) using three techniques inclu ...rack (automatically), measuring from Pitch track manually, and by lookin ...highlighting one cycle. Before measuring the fundamental frequency, we nee ...containing a vowel sound (e.g., [æ]). Let's learn all of the three methods one by

1. Displaying the pitch track and allowing Praat to measure the pitch automati
   - Display the pitch track: Pitch > Show pitch.
   - Place your cursor in the middle – a stable portion of the vowel.
   - Go to Pitch > Get pitch – a box will appear with the pitch value in it (r

2. Displaying the pitch track and measuring pitch manually:
   - Display the pitch track - Pitch > Show pitch.
   - Click on the blue pitch track in the middle of the vowel.
   - A red horizontal bar should appear with the pitch value (in dark blue) window (take the measurement from here).

3. By looking at the waveform (top of the display):

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h is calculated by looking at the wa

- Zoom into a small piece of the waveform in the middle highlighting one complete cycle and noting the time ass waveform).

Topic-188: Measuring the Harmonics

Harmonics are the multiple integers of the fundamenta of vocal fold vibration (complex wave). We need the 'narrow (which we can set by fixing the spectrum setting at 0.025). B first three harmonics, we will go to the H10 (H1, H2, H3 – H measurement already taken. It is important to note down that w complex wave, consisting of the fundamental frequency (whic other higher frequencies, called harmonics. As already mention narrow-band spectrogram, which is more precise along the frec pectrogram. Let's now take the harmonics:

. Display a narrow-band spectrogram:

- Go to: Spectrum > Spectrogram settings.
- Change the window length to 0.025s – the defau spectrogram) - this changes the spectrogram dramatical
- Looking at each vowel, notice the grey horizontal band vowel, measure the frequencies of the first 3 harmonics
- Click on the center (horizontally) of each harmonic in t
- A red horizontal bar should appear with the frequency

pic-189: Measuring the Formants

Formants are the overtone resonances. Acoustically, ir important. By now, we know that the narrow bands s nonics whereas we need the wide bands for measuring acteristics of sonorant speech sounds – vowels). On spectr esponds to loudness; i.e. the darkest harmonics are the lified harmonics form the formants that are characteristi sure the first and second formants (F1 and F2) from th niques outlined below and note down your measurements:

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USING PRAAT-III

section, the students will be able to:

vel properties
and ANALYSE features such as intrinsic pitch and spectral make-up (f

Properties

ng the acoustics of vowels in this session, we need to record vowels a
t vowels from American English (given in your book (Chapter 8) ar
now, you should know how to record them). These vowels are: heed
and who'd. When you are done with the recording, get ready fo
gs: intrinsic pitch, spectral make up (formants) and plotting them in
m to your Word document). Now, record yourself saying the word
in the Edit window, and make sure you can clearly see the vowel
hem, you can go back to the previous labs and learn it again. While d
your worksheet.

Pitch

ight vowels recorded in the last session and measure the pitch (F0) f
tch by using any of the three ways already discussed in the To
F0) in each of the vowels (if your pitch values seem strange, tr
for confirmation; remember that measuring the waveform is the s
ents. There is one more way to confirm your pitch measurement b
gives the component frequencies and their amplitudes).

on of the vowel (70-80ms).
iew spectral slice.
irst (big) peak = H1 = F0 (Ignore any small spikes at the beginni
ote down the frequency of this peak at the top of the vertical bar).

l pitch values and plot the pitch of each vowel on your excel shee
a scale that allows you to spread out your measurements as much
from the excel sheet and export to Word document and give the fi

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In order to ... up of the vowels, we need to
mants i.e., F1 and F2) of the eight vowels already recorded. Ca
vel. This you can do by using the automatic formant tracking o
gment than that of Praat). Having taken the values for all of the
a chart. Remember that we need the default wide-band spectro
ormants (F1 and F2) of each vowel. You can also use Praat's au
ou want. Also note and try to answer the following question
and lectures on acoustics):
lo formants (F1 and F2) differ across vowels?
does F1 seem to correspond to, in terms of articulation?
about F2? To what does it correspond?

94. Plotting Vowels on Chart

Finally, we are at the last step related to spectral make-up of
e measurement of the first two formants and we are going to plot
preadsheet. By putting F1 and F2 in separate columns, write th
t vowels (giving vowels in the first column, the difference be
and F1 in the third). After putting the data in Excel sheet, we w
preadsheet. Further in order to make it corresponding with the re
erse the values for both formants (on both axis – Y and X). Now
t corner. Watch the video and you will find how F1 is inversely
difference between F2 and F1 to the frontness of the vowels. Or
ord document and give it the number and title accordingly.

r readings on this section:

ter 3 (Lab 3) of **Acoustic Phonetics Lab Manual** (developed by
ersity of Victoria)

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USING PRAAT - IV

t the end of this section, the students will be able to:

EXPLAIN sonorants in terms of acoustics.
RECOGNIZE AND ANALYSE the features of nasals, glides and liqu

pic-195: Sonorants and Their Formants

Sonorants are vowel-like sounds (nasals and glides). These soun
y have formants (remember their acoustic correlates). But they are dif
ierally have lower amplitude; therefore, they behave like consonants.
our experimentation on sonorant sounds /ama/ - /ana/ - /aɲa/ - /wi
uences, now start exploring the features of these sounds like the mea
to compare them with vowels. Do you see any systematic patterns? T
ving to the next topic.

ic-196: Nasal Formants

Formants for nasal sounds are also important for acoustic analysis
F3) formants of nasals from the file (use the already learnt way of n
nasals have very distinctive waveforms (different than that of vowels
nti-formants (bands of frequencies damped) and formant transition
urement, try to answer the following questions:

. Are there any systematic patterns across nasals?
. Is there one formant with a similar frequency for all places of artic
. Is there one formant that has much higher amplitude than the othe
Do you see any overall differences between the nasals on the one

-197: Glide and Their Formants

Glides are also the sonorants (vowel-like) sounds as they have s
from our recorded file and take the first three formants (F1, F2 a
s for glides (both for /w/ and /j/) and explore their acoustic correlat
sounds (the midpoint of [w] and [j]). Analyze that how similar is
owels and nasals. Draw lines to indicate F1, F2, F3 and compare w
answer the following questions:

Which vowel does each glide resemble?
Are there any differences between the contours, i.e. between the tr
What do you think may cause these differences (in terms of articu

er readings on this section:

ter 4 (Lab 4) of **Acoustic Phonetics Lab Manual** (developed by So
ersity of Victoria)

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ıation of different features (e.g., /d/ as a phoneme and its features – ıtral) in a binary (+ -) order which is an important component of ıalysis may also include aspects of the target language as the part of learning point of view). Remember that we need to include three astive function (how it is different), descriptive function (what it is) broader classes of sounds). Features may also be studied further as a their role as language specific sub sets.

s

and phonology, the studies of sounds include various latest nputer software that are used under carefully designed lab ıspect of the application of the latest technology by going beyond the sophisticated phonetic labs in order to discover the hidden aspects of ns such as 'How speech is produced and processed?' are the focus of ıe speech chain as the beginning of experimental phonetics as Coach). The latest trends under experimental phonetics include brain processing (by using the latest equipment – many special instruments errors, neurolinguistics and the topics related to the developments ysis and synthesis.

many other major languages such as Urdu, Punjabi and Pashto, fo d as potential areas of research in the domain of phonetics an lude comparisons and contrasts (similarities and differences) amon anguage(s) – e.g., differences at phonetic and phonological levels an mental features – in different varieties. Differences among accents ed in this course under various headings (such as vowels in oth lish dialectology has already been explored by a number of studi phic differences (in the recognition of various forms of English – ell-known data gathering techniques used in field linguistics (such eties) like variations paradigm.. Field workers are particularly train tise are developed for large scale studies related to the studies

THE PEDAGOGY OF PHONETICS AND PHONOL

e end of this section, the students will be able to:

DENTIFY the best method for the teaching of P&P to Pakistani students.

-205: The Relationship Between ELT and P&P

This course has many important pedagogical implications for the teacher logy. Phonetics and phonology is an integral part of English Language Te ıg of phonetics and phonology further needs to be integrated in the teachir e, the teachers are expected to work on their skills related to the pronuncia ze their students related to the topic – they may use their self-initiated procedu nological contrastive analysis (CA) (e.g., of their mother tongues and Englis gmental levels for enhancing their skills and completing their researches. The part in the phonology based ELT activities from the TESOL Home Page (a ate in the English Language Teaching Reforms (ELTR) projects of the ssion (HEC) of Pakistan and activities planned and sponsored by British Co are also expected to be the part of these platforms through their social groups nities.

06: Developing Relevant Material

Developing relevant material for the teaching of phonetics and phonology is a teachers of English language. Remember the specific needs of ELT activities ore already developed material available online from various sources (such r teacher resource centers); however, you must also be able to develop you lly required by your students). For example, you can develop your mat ation teaching to the learners of English. You can incorporate material relate tion of the audio (listening) based activities – by involving students on using etic dictionaries) in the classroom.

Movies and documentaries (such as from BBC - CNN - National Geographic very effective resources for the teaching of pronunciation. Finally, the rea and writing interaction from everyday language may also yield tremendous r development should always be the enhancement of the proficiency level of stu

7: Conducting Classroom Research for ELT

s a part of the pedagogy of the phonetics and phonology, the teachers are ex earch which is an important aspect of ELT with significant pedagogical in n this context means focusing on the issues faced by ELT instructors an

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Topic-208: Making Research Accessible to Teachers

Good teachers are expected to be active researchers and therefore busy in updating themselves bout the latest researches and teaching methodologies around the world. It is also a pedagogical hallenge for teachers to keep themselves updated by exploring pedagogical and technological challenges or ELT experts (in their own contexts and internationally). For example, the aspects of Task Based earning and Teaching (TBLT) as a golden method for second language acquisition (SLA) may be ffective in Pakistani context if explored by ELT practitioners. Teachers as the agents of change and they must be reading research studies and carry out research by and explore their issues and solutions. A good way is to keep reading teachers' digests and journals and participate in the online discussions by teaching associations.

Topic-209: Facilitating Action Research

teachers are expected to facilitate action research which is the most ession. For example, the phonetics of phonological speech own practices) may lead to a very positive ies such as learners'

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